Mathematical Foundations and Architecture of the Perpetual Swap Ruin Engine

Technical Documentation and Parameter Guide for `core_math_v2.py`

Quantitative Derivatives Research

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Abstract

This document provides an accessible, step-by-step mathematical explanation of the computational core implemented in `core_math_v2.py`. The engine calculates the probability of forced liquidation (trader ruin) for perpetual futures contracts in cryptocurrency markets. We explain every financial mechanism and mathematical principle using plain English intuition, explicit numbered formulas, detailed parameter catalogs, and practical limitations. Key topics include tiered maintenance margin schedules, liquidation fees, adverse execution slippage, coin-margined (inverse) versus linear settlement, discrete barrier corrections using the Broadie-Glasserman-Kou method, intra-step Brownian bridge crossings, Merton jump-diffusion with Heston stochastic volatility, regime-switching funding rates, Bayesian drift shrinkage, and risk-constrained Half-Kelly leverage optimization.

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1 Introduction and General Overview

In cryptocurrency trading, a perpetual futures contract ("perp") is a derivative agreement with three unique properties:

1. **No Expiration Date:** Traders can keep a position open indefinitely, provided they meet exchange margin requirements.
2. **Periodic Funding Rate:** Cash transfers occur regularly (typically every 8 hours) between longs and shorts to anchor the contract price to the spot index price.
3. **Forced Liquidation Boundary:** If paper losses and funding fees reduce a trader's posted collateral below the exchange's required maintenance threshold, the exchange closes the position immediately.

The mission of `core_math_v2.py` is to answer one vital question: **What is the exact mathematical probability that an open position will hit its liquidation price before the trade duration ends?**

2 Principle 1: Standard Normal Cumulative Distribution Function

2.1 Simple Language Explanation

In a standard bell curve (Gaussian distribution), most random shocks are small and centered near zero, while massive shocks are rare. The cumulative distribution function $\Phi(x)$ answers: "What is the chance that a standard normal shock is less than or equal to x?" If $x = 0$, the answer is exactly 0.5 (50%), because half of all shocks are negative and half are positive.

2.2 Mathematical Formulation

The cumulative distribution function $\Phi(x)$ for a standard normal random variable $Z \sim \mathcal{N}(0, 1)$ is the integral of its probability density function:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{u^2}{2}}\, du \tag{1}$$

Because this integral has no elementary closed-form solution, the code computes it using the standard Gauss error function $\mathrm{erf}(z)$:

$$\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \int_{0}^{z} e^{-t^2}\, dt \tag{2}$$

Substituting $u = t\sqrt{2}$ yields the exact formula executed inside `norm_cdf`:

$$\Phi(x) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)\right] \tag{3}$$

2.3 Parameter Reference

- `x` (`float`): The z-score representing how many standard deviations the test value is away from the zero mean ($-\infty < x < \infty$).

2.4 Limitations

- **Floating-Point Underflow:** For extreme tail values ($|x| > 8.0$), standard 64-bit floating-point math rounds $\Phi(x)$ directly to 0.0 or 1.0, obscuring deep-tail probabilities.
- **Symmetry Assumption:** Equation (1) assumes returns are perfectly symmetrical. In real crypto markets, price crashes are steeper than rallies (negative skewness).

3 Principle 2: Mechanical Liquidation Prices with Tiered Margins, Fees, and Margin Modes

3.1 Simple Language Explanation

Your liquidation price is the price level where paper losses wipe out your initial margin. When price hits this boundary, the exchange seizes and closes your position to protect its own solvency.

- **Tiered Margin:** Large positions require higher safety collateral than small positions. A retail position may require 0.5% maintenance margin, but a large institutional position might require 2.0% or 5.0%.
- **Liquidation Fee:** The exchange charges an extra penalty when you are liquidated to fund its insurance pool.
- **Execution Slippage:** During a sudden flash crash, order books thin out. Your liquidation order is filled at worse prices than expected. The slippage buffer accounts for this by shifting the liquidation trigger closer to the current price.
- **Linear vs. Inverse Contracts:** In a linear contract (USDT/USDC margined), collateral is stable cash. In an inverse contract (coin-margined), collateral is the volatile cryptocurrency itself (e.g., BTC). When price crashes, you suffer a double loss: your trade is losing, and the coin backing your trade is losing purchasing power at the exact same time. Hence, inverse contracts get liquidated much faster.

3.2 Mathematical Formulation

3.2.1 1. Tiered Margin Schedule and Effective Margin Rate

Let posted margin be $C > 0$, leverage be $L > 1.0$, and entry price be P_0. The nominal position notional value is:

$$V_0 = C \cdot L \tag{4}$$

The base maintenance margin rate $\text{MMR}(V_0)$ is resolved via a piecewise-constant institutional bracket schedule:

$$\text{MMR}(V_0) = \begin{cases} 0.005 \quad (0.50\%) & \text{if } |V_0| \le 50{,}000 \\ 0.010 \quad (1.00\%) & \text{if } 50{,}000 < |V_0| \le 250{,}000 \\ 0.020 \quad (2.00\%) & \text{if } 250{,}000 < |V_0| \le 1{,}000{,}000 \\ 0.050 \quad (5.00\%) & \text{if } |V_0| > 1{,}000{,}000 \end{cases} \tag{5}$$

Adding the liquidation clearance fee rate f_{liq} gives the effective adjusted maintenance margin rate:

$$\text{MMR}_{\text{adj}} = \text{MMR}(V_0) + f_{\text{liq}} \tag{6}$$

The initial margin rate required to open the trade is:

$$\text{IMR}_0 = \frac{1}{L} \tag{7}$$

3.2.2 2. Linear Contracts (USDT / USDC Margined)

Account equity $E(P)$ at market price P for contract quantity $Q = V_0/P_0$ is:

$$E_{\text{long}}(P) = C + Q(P - P_0) = C + \frac{V_0}{P_0}(P - P_0) \tag{8}$$

$$E_{\text{short}}(P) = C - Q(P - P_0) = C - \frac{V_0}{P_0}(P - P_0) \tag{9}$$

Liquidation triggers when equity drops to the maintenance threshold $E(P_{\text{liq}}) = V_0 \cdot \text{MMR}_{\text{adj}}$:

$$P^{\text{long}}_{\text{liq,linear}} = P_0 \left(1 - \frac{1}{L} + \text{MMR}_{\text{adj}}\right) \tag{10}$$

$$P^{\text{short}}_{\text{liq,linear}} = P_0 \left(1 + \frac{1}{L} - \text{MMR}_{\text{adj}}\right) \tag{11}$$

3.2.3 3. Inverse Contracts (Coin-Margined)

For coin-margined positions, collateral is denominated in the underlying cryptocurrency:

$$E^{\text{inv}}_{\text{long}}(P) = \frac{C_{\text{usd}}}{P} + V_0 \left(\frac{1}{P_0} - \frac{1}{P}\right) \tag{12}$$

Setting $E^{\text{inv}}_{\text{long}}(P_{\text{liq}}) = (V_0/P_{\text{liq}}) \cdot \text{MMR}_{\text{adj}}$ gives the non-linear hyperbolic liquidation price:

$$P^{\text{long}}_{\text{liq,inverse}} = \frac{P_0}{1 + \frac{1}{L} - \text{MMR}_{\text{adj}}} \tag{13}$$

$$P^{\text{short}}_{\text{liq,inverse}} = \frac{P_0}{1 - \frac{1}{L} + \text{MMR}_{\text{adj}}} \tag{14}$$

3.2.4 4. Adverse Slippage Adjustment

To account for shallow order books during flash crashes, an adverse execution slippage buffer f_{slip} shifts the barrier closer to entry:

$$P^{\text{long}}_{\text{liq,eff}} = P^{\text{long}}_{\text{liq}} \cdot (1 + f_{\text{slip}}) \tag{15}$$

$$P^{\text{short}}_{\text{liq,eff}} = P^{\text{short}}_{\text{liq}} \cdot \max\left(10^{-6},\ 1 - f_{\text{slip}}\right) \tag{16}$$

3.3 Parameter Reference

- `entry_price` (P_0): Execution price at trade entry.
- `leverage` (L): Leverage multiple ($L > 1.0$).
- `position_type`: Direction of trade, `"long"` or `"short"`.
- `collateral` (C): Margin posted in USD, used to look up the correct tier.
- `maintenance_margin_rate`: Optional manual override for MMR.

- `liquidation_fee_rate` (f_{liq}): Mandatory exchange clearance fee (default 0.1%–0.15%).
- `slippage_rate` (f_{slip}): Adverse execution buffer in thin order books.
- `margin_mode`: Settlement structure, `"linear"` or `"inverse"`.
- `custom_tiers`: List of tier thresholds and rates.

3.4 Limitations

- **Fixed Slippage Rate:** Real slippage depends dynamically on the depth of the book at the moment of liquidation.
- **Mark Price vs. Last Price:** Exchanges trigger liquidations using an index-derived Mark Price rather than the last traded price.

4 Principle 3: First-Passage Time Ruin with Broadie-Glasserman-Kou Correction

4.1 Simple Language Explanation

Classic textbook models assume that prices move in an uninterrupted, infinitely smooth stream that is watched every microsecond. In reality, exchanges monitor margin and mark prices at discrete intervals (e.g., every second or upon trade events). Because monitoring happens at discrete steps, the chance of catching a brief spike is slightly lower than under continuous monitoring.

The **Broadie-Glasserman-Kou (BGK)** correction shifts the continuous barrier just enough to compensate for discrete checking, giving an exact match with real exchange behavior.

4.2 Mathematical Formulation

4.2.1 1. Continuous-Time Price Dynamics

Under Geometric Brownian Motion (GBM), the log price ratio $X_t = \ln(S_t/S_0)$ follows an arithmetic Brownian motion with annualized drift m and volatility σ:

$$dX_t = m\,dt + \sigma\,dW_t, \quad m = \mu - \frac{1}{2}\sigma^2 \tag{17}$$

4.2.2 2. The Broadie-Glasserman-Kou (1997) Correction

When prices are checked periodically at intervals Δt, the continuous barrier B is shifted away from spot by the BGK constant β:

$$\beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} \approx 0.5825971579 \tag{18}$$

$$\text{Shift Factor} = \exp\left(\beta \cdot \sigma\sqrt{\Delta t}\right) \tag{19}$$

For a long position (lower barrier), discrete monitoring shifts the barrier downward (further from S_0, harder to breach):

$$B_{\text{adj}}^{\text{long}} = B \cdot \exp\left(-\beta \cdot \sigma\sqrt{\Delta t}\right) = \frac{B}{\exp\left(\beta \cdot \sigma\sqrt{\Delta t}\right)} \tag{20}$$

For a short position (upper barrier), it shifts upward:

$$B_{\text{adj}}^{\text{short}} = B \cdot \exp\left(\beta \cdot \sigma\sqrt{\Delta t}\right) \tag{21}$$

4.2.3 3. Analytical Ruin Probability Formula

Over holding horizon $T = T_{\text{days}}/365$, the probability that the running minimum/maximum hits the shifted barrier is derived from the Bachelier-Harrison reflection principle. For a long position ($B_{\text{adj}}^{\text{long}} < S_0$):

$$h = \ln\left(\frac{B_{\text{adj}}^{\text{long}}}{S_0}\right) < 0, \qquad d_1 = \frac{h - mT}{\sigma\sqrt{T}}, \qquad d_2 = \frac{h + mT}{\sigma\sqrt{T}} \tag{22}$$

$$\text{scale} = \frac{2mh}{\sigma^2}, \qquad \mathbb{P}(\tau \le T) = \Phi(d_1) + e^{\text{scale}}\Phi(d_2) \tag{23}$$

For a short position ($B_{\text{adj}}^{\text{short}} > S_0$):

$$h = \ln\left(\frac{S_0}{B_{\text{adj}}^{\text{short}}}\right) < 0, \qquad d_1 = \frac{h + mT}{\sigma\sqrt{T}}, \qquad d_2 = \frac{h - mT}{\sigma\sqrt{T}} \tag{24}$$

$$\text{scale} = -\frac{2mh}{\sigma^2}, \qquad \mathbb{P}(\tau \le T) = \Phi(d_1) + e^{\text{scale}}\Phi(d_2) \tag{25}$$

4.3 Parameter Reference

- `S0`: Initial asset spot price.
- `B`: Liquidation barrier price.
- `sigma` (σ): Annualized continuous price volatility.
- `T_days`: Holding period in calendar days.
- `position_type`: `"long"` or `"short"`.
- `mu` (μ): Annualized drift rate (default `0.0`).
- `steps_per_day`: Discrete checks per day used to calculate Δt.

4.4 Limitations

- **Constant Volatility Assumption:** Equations (23) and (25) assume fixed volatility σ, omitting volatility clustering.
- **Static Barrier:** Analytical reflection formulas cannot directly track dynamic funding fee decay, which moves the barrier continuously over time.

5 Principle 4: Continuous Intra-Step Crossing via Brownian Bridge

5.1 Simple Language Explanation

In a Monte Carlo simulation with hourly steps, suppose the price was safe at 1:00 PM and safe again at 2:00 PM. Does that prove the trader was safe? Not necessarily! At 1:25 PM, a flash dip could have touched the liquidation line and bounced right back.

Checking prices only at hourly endpoints underestimates liquidations. A **Brownian Bridge** computes the exact probability that the continuous price path dipped below the barrier in-between two safe observation points, ensuring no intra-step liquidation is missed.

5.2 Mathematical Formulation

Given that the asset price starts at S_k at time t and ends at S_{k+1} at time $t + \Delta t$, where both endpoints are strictly on the safe side of barrier B, the probability that the continuous path touched or crossed B during the interval is:

$$p_{\text{intra}}^{\text{long}} = \exp\left(-\frac{2\ln(S_k/B)\ln(S_{k+1}/B)}{\sigma_k^2 \Delta t}\right) \tag{26}$$

$$p_{\text{intra}}^{\text{short}} = \exp\left(-\frac{2\ln(B/S_k)\ln(B/S_{k+1})}{\sigma_k^2 \Delta t}\right) \tag{27}$$

The engine draws a uniform random number $U \sim \mathcal{U}(0,1)$. If $U < p_{\text{intra}}$, the path is marked as liquidated during that step.

5.3 Parameter Reference

- `S_start`, `S_end`: Price vectors at the start and end of the time step.
- `barrier`: Liquidation boundary vector across the interval.
- `sigma`: Local volatility of the step.
- `dt`: Time step duration in years (Δt).

5.4 Limitations

- **Pure Diffusion Assumption:** The Brownian Bridge formula strictly models Gaussian diffusion. If a discontinuous jump occurs inside the interval, the crossing formula must be adapted.

6 Principle 5: Merton Jump-Diffusion, Heston Volatility, and Regime-Switching Funding

6.1 Simple Language Explanation

Real crypto markets deviate from smooth models in four critical ways:

1. **Price Jumps (Merton):** Crypto prices can gap down 10% in seconds due to breaking news. Merton's model adds unexpected Poisson jumps on top of ordinary volatility.

2. **Exploding Volatility (Heston):** When markets crash, volatility explodes (the leverage effect). Heston's model treats volatility as an evolving random process negatively correlated with price.
3. **Funding Rate Regimes:** When the market is bullish, funding is positive. When the market plunges into panic (e.g., a $> 10\%$ crash), funding flips negative as shorts pay longs.
4. **Coin-Margined Collateral Bleed:** For inverse contracts, funding fees are deducted in actual cryptocurrency coins. As the price falls, remaining coin collateral loses USD purchasing power simultaneously.

6.2 Mathematical Formulation

6.2.1 1. Heston Stochastic Volatility (Full-Truncation CIR Process)

Instantaneous variance $v_t = \sigma_t^2$ evolves under a Cox-Ingersoll-Ross (CIR) process anchored to long-run target $\theta_v = v_0 = \sigma_0^2$:

$$dv_t = \kappa_v(\theta_v - v_t)\,dt + \sigma_v\sqrt{v_t}\,dW_t^{(v)} \tag{28}$$

where variance shocks correlate negatively with asset price shocks ($\rho_{Sv} < 0$):

$$dW_t^{(v)} = \rho_{Sv}\,dW_t^{(S)} + \sqrt{1-\rho_{Sv}^2}\,dZ_t^{(v)}, \quad Z_t^{(v)} \perp W_t^{(S)} \tag{29}$$

To prevent negative variance when the Feller condition ($2\kappa_v\theta_v > \sigma_v^2$) is violated, the engine applies a Full-Truncation Euler discretization across time step Δt:

$$\tilde{v}_k = \max(0, v_k), \quad v_{k+1} = \max\left(10^{-6},\ v_k + \kappa_v(\theta_v - \tilde{v}_k)\Delta t + \sigma_v\sqrt{\tilde{v}_k}\sqrt{\Delta t}\,Z_k^{(v)}\right) \tag{30}$$

6.2.2 2. Merton Jump-Diffusion Process and Discrete Log-Euler Step

Asset prices evolve under local volatility $\sigma_t = \sqrt{v_t}$ and compound Poisson jumps:

$$\frac{dS_t}{S_{t^-}} = (\mu - \lambda\kappa)\,dt + \sigma_t\,dW_t^{(S)} + (e^J - 1)\,dN_t \tag{31}$$

where N_t has intensity λ, $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$, and κ is the compensator:

$$\kappa = \mathbb{E}[e^J - 1] = \exp\left(\mu_J + \frac{1}{2}\sigma_J^2\right) - 1 \tag{32}$$

Over discrete interval Δt, let $N_k \sim \text{Poisson}(\lambda\Delta t)$ denote the jump count. The discrete log-price increment is:

$$\Delta \ln S_k = \left(\mu - \frac{1}{2}\sigma_k^2 - \lambda\kappa\right)\Delta t + \sigma_k\sqrt{\Delta t}\,Z_k^{(S)} + N_k\mu_J + \sqrt{N_k}\sigma_J Z_k^{(J)} \tag{33}$$

6.2.3 3. Regime-Switching Stochastic Funding Rates

Funding rates follow an Ornstein-Uhlenbeck (OU) process where the mean-reversion target switches depending on whether price has dropped more than 10%:

$$r_{k+1} = r_k + \theta\left(\bar{r}(S_k) - r_k\right)\Delta t + \sigma_r\sqrt{\Delta t}\,Z_k^{(r)} \tag{34}$$

$$\bar{r}(S_k) = \begin{cases} \bar{r}_{\text{bull}} & \text{if } S_k/S_0 \geq 0.90 \\ \bar{r}_{\text{bear}} & \text{if } S_k/S_0 < 0.90 \end{cases} \tag{35}$$

where $Z_k^{(r)} = \rho_{Sf}Z_k^{(S)} + \sqrt{1-\rho_{Sf}^2}Z_k^{(\perp)}$.

6.2.4 4. Collateral Dynamics and Dynamic Liquidation Barrier Evolution

In linear contracts (USDT/USDC margined), USD collateral bleeds at each step k:

$$C_k^{\text{USD}} = \max\left(0,\ C_{k-1}^{\text{USD}} \mp V_0 \cdot r_{k-1} \cdot \Delta t\right), \quad \text{IMR}_k^{\text{lin}} = \max\left(10^{-4},\ \frac{C_k^{\text{USD}}}{V_0}\right) \tag{36}$$

where $-$ applies to longs and $+$ applies to shorts. In inverse (coin-margined) contracts, funding is deducted in units of the cryptocurrency ($C_0^{\text{coin}} = C_0/S_0$):

$$C_k^{\text{coin}} = \max\left(0,\ C_{k-1}^{\text{coin}} \mp \left(\frac{V_0}{S_k}\right) r_{k-1}\Delta t\right), \quad \text{IMR}_k^{\text{inv}} = \max\left(10^{-4},\ \frac{C_k^{\text{coin}} \cdot S_0}{V_0}\right) \tag{37}$$

Anchoring $\text{IMR}_k^{\text{inv}}$ to entry price S_0 ensures that mark-to-market drops are not double-counted in the mechanical barrier. Substituting IMR_k and effective maintenance margin $\text{MMR}_{\text{adj}} = \text{MMR}(V_0) + f_{\text{liq}}$ yields the dynamic barrier B_k:

$$B_k^{\text{lin,long}} = S_0\left(1 - \text{IMR}_k^{\text{lin}} + \text{MMR}_{\text{adj}}\right)(1 + f_{\text{slip}}) \tag{38}$$

$$B_k^{\text{lin,short}} = S_0\left(1 + \text{IMR}_k^{\text{lin}} - \text{MMR}_{\text{adj}}\right)\max(10^{-6}, 1 - f_{\text{slip}}) \tag{39}$$

$$B_k^{\text{inv,long}} = \frac{S_0}{\max\left(10^{-6},\ 1 + \text{IMR}_k^{\text{inv}} - \text{MMR}_{\text{adj}}\right)}(1 + f_{\text{slip}}) \tag{40}$$

$$B_k^{\text{inv,short}} = \frac{S_0}{\max\left(10^{-6},\ 1 - \text{IMR}_k^{\text{inv}} + \text{MMR}_{\text{adj}}\right)}\max(10^{-6}, 1 - f_{\text{slip}}) \tag{41}$$

6.3 Parameter Reference

- `heston_kappa` (κ_v), `heston_vol_of_vol` (σ_v): Mean-reversion speed and vol-of-vol.
- `heston_price_vol_corr` (ρ_{Sv}): Asset-volatility correlation (typically -0.60).
- `jump_intensity_lambda` (λ): Poisson jump frequency per year.
- `jump_mean_mu` (μ_J), `jump_std_sigma` (σ_J): Log jump size distribution.
- `funding_ou_theta` (θ), `funding_ou_sigma` (σ_r): OU funding parameters.
- `funding_price_corr` (ρ_{Sf}): Correlation between price shocks and funding fees.
- `funding_bear_rate` ($\bar{r}_{\text{bear}}$): Mean funding rate during panics (default -5%).
- `num_paths`, `steps_per_day`: Monte Carlo path count ($5{,}000$) and daily steps (24).

6.4 Limitations

- **Calibration Complexity:** Calibrating Heston and jump parameters requires liquid options chains, which are unavailable for smaller altcoins.
- **Monte Carlo Statistical Error:** Monte Carlo precision converges at rate $\mathcal{O}(1/\sqrt{M})$.

7 Principle 6: Bivariate Ruin Risk Heatmap Grid

7.1 Simple Language Explanation

Rather than analyzing a single trade, this module sweeps across 25 leverage levels and 25 volatility levels simultaneously (625 total grid intersections) to visually map safe zones versus rapid liquidation cliffs.

7.2 Mathematical Formulation

The parameter space $\Sigma \times \Lambda$ is discretized into a uniform $G \times G$ Cartesian grid:

$$\sigma_i = \sigma_{\text{min}} + \frac{i-1}{G-1}(\sigma_{\text{max}} - \sigma_{\text{min}}), \qquad i \in \{1, \ldots, G\} \tag{42}$$

$$L_j = L_{\text{min}} + \frac{j-1}{G-1}(L_{\text{max}} - L_{\text{min}}), \qquad j \in \{1, \ldots, G\} \tag{43}$$

At each intersection (σ_i, L_j), the analytical ruin probability is evaluated:

$$M_{i,j} = \text{analytical_ruin_probability}\left(P_0, P_{\text{liq}}(P_0, L_j), \sigma_i, T, \text{pos}, \mu\right) \tag{44}$$

7.3 Parameter Reference

- `sigma_min`, `sigma_max`: Volatility bounds (default 20% to 140%).
- `leverage_min`, `leverage_max`: Leverage bounds (default $2\times$ to $50\times$).
- `grid_size`: Resolution per axis (default 25, producing 625 evaluations).
- `duration_days`: Holding period in days.

8 Principle 7: Risk-Constrained Half-Kelly Leverage Optimization

8.1 Simple Language Explanation

- **The Kelly Criterion:** A formula for calculating the trade size that maximizes long-term compound wealth growth. However, standard Kelly assumes you cannot be wiped out by an absorbing barrier. In leverage trading, pure Kelly sizing often leads directly to liquidation.
- **Directional Drift Alignment:** Expected drift must align with trade direction: a positive drift benefits longs, whereas a negative drift benefits shorts.
- **Half-Kelly:** We scale the raw Kelly fraction down by 50%. This preserves three-quarters of the maximum growth rate while reducing portfolio volatility and drawdown by half.
- **Ruin-Probability Cap:** A numerical bisection search finds the highest leverage where the probability of liquidation remains strictly under a safety ceiling (e.g., $\leq 1\%$). The recommendation is the smaller of Half-Kelly or the safe liquidation limit.

8.2 Mathematical Formulation

8.2.1 1. Directional Merton-Bayes Shrunk Drift

Let directional drift aligned with trade orientation be:

$$\mu_{\text{dir}} = \begin{cases} +\mu & \text{if position is long} \\ -\mu & \text{if position is short} \end{cases} \tag{45}$$

When raw historical drift is supplied over lookback $T_{\text{sample}} = N_{\text{days}}/365$, Bayesian shrinkage regularizes the estimate using prior variance $\sigma^2_{\text{prior}} = (0.15)^2$:

$$w = \frac{\sigma^2_{\text{prior}}}{\sigma^2_{\text{prior}} + \frac{\sigma^2}{T_{\text{sample}}}}, \quad \mu_{\text{eff}} = w \cdot \mu_{\text{dir}} \tag{46}$$

(If input drift μ has already been regularized by `market_data_v2.py`, $\mu_{\text{eff}} = \mu_{\text{dir}}$ is used directly without duplicate shrinkage.)

8.2.2 2. Fractional Half-Kelly Leverage

For $\mu_{\text{eff}} > 0$, the growth-optimal fractional Kelly leverage with multiplier $c = 0.5$ is:

$$f^*_{\text{half}} = \max\left(1.0,\ c \cdot \frac{\mu_{\text{eff}}}{\sigma^2}\right) \tag{47}$$

If $\mu_{\text{eff}} \leq 0$, the unconstrained Kelly leverage defaults to the minimum baseline $f^*_{\text{half}} = 1.0$.

8.2.3 3. Bisection Optimization for Ruin-Constrained Leverage

The algorithm searches for the highest leverage L_{safe} satisfying the first-passage ruin ceiling $\alpha_{\max}$ under conservative zero drift ($\mu = 0$):

$$L_{\text{safe}} = \sup\left\{L \in [1.01, L_{\max}] : \mathbb{P}_{\text{ruin}}(L; P_0, \sigma, T, \mu = 0) \leq \alpha_{\max}\right\} \tag{48}$$

Solved via 40 iterations of binary bisection, the final recommended leverage is:

$$L^* = \min\left(f^*_{\text{half}},\ L_{\text{safe}}\right) \tag{49}$$

8.3 Parameter Reference

- `mu`: Annualized expected return of the asset.
- `sigma`: Annualized price volatility.
- `max_ruin_prob` ($\alpha_{\max}$): Maximum permissible liquidation probability (default 0.01, or 1%).
- `max_allowed_leverage` ($L_{\max}$): Exchange maximum leverage limit (default 50.0).
- `use_half_kelly`: Toggle for fractional Half-Kelly scaling ($c = 0.5$).
- `apply_shrinkage`: Toggle to guard against duplicate Bayesian shrinkage.

8.4 Limitations

- **Drift Estimation Sensitivity:** Real drift is notoriously difficult to measure. Setting μ slightly too high can cause Kelly formulas to recommend dangerously aggressive positions.

9 Principle 8: Historical Black Swan and Short Squeeze Stress-Testing

9.1 Simple Language Explanation

Instead of trusting equations alone, we conduct a real-world flight test: "If you opened this exact trade with your chosen leverage during March 2020 Black Thursday, the May 2021 mining ban, the November 2022 FTX collapse, or the July 2021 short squeeze, would your account have survived? At what exact hour would you have been liquidated?"

9.2 Mathematical Formulation

The function maps a historical hourly price multiplier trajectory m_t over duration H hours:

$$S_t^{\text{long}} = P_0 \cdot m_t, \quad t \in \{0, 1, \ldots, H\} \tag{50}$$

For short positions, authentic short squeeze data is applied directly, or the crash trajectory is inverted:

$$S_t^{\text{short}} = P_0 \cdot (2.0 - m_t) \tag{51}$$

The first hour of liquidation τ_{breach} is identified by:

$$\tau_{\text{breach}} = \min \left\{ t : S_t \leq P_{\text{liq}}^{\text{long}} \text{ or } S_t \geq P_{\text{liq}}^{\text{short}} \right\} \tag{52}$$

The maximum adverse price excursion (Maximum Drawdown) is:

$$\text{MDD}_{\text{long}} = \frac{\min_t S_t}{P_0} - 1, \quad \text{MDD}_{\text{short}} = \frac{\max_t S_t}{P_0} - 1 \tag{53}$$

9.3 Parameter Reference

- `scenario_key`: Key identifying the event (e.g., `"black_thursday_2020"`, `"short_squeeze_2021"`).
- `entry_price`, `leverage`, `position_type`, `margin_mode`: Trade specifications.

9.4 Limitations

- **Hourly Sample Granularity:** Historical trajectories are recorded at 1-hour intervals, which can overlook multi-second flash wicks.

10 Master Parameter Catalog for the Engine

Table 1 provides a consolidated reference for all primary parameters in `core_math_v2.py`.

Table 1: Consolidated Engine Parameter Catalog

Parameter Name	Symbol	Type	Default	Operational and Financial Role
`entry_price` / `S0`	P_0, S_0	`float`	—	Reference entry spot price of the asset.
`leverage`	L	`float`	—	Nominal trade leverage multiplier ($L > 1.0$).
`position_type`	dir	`str`	`"long"`	Directional orientation: `"long"` or `"short"`.
`collateral`	C	`float`	`None`	Initial posted margin capital in USD.
`maintenance_margin_rate`	MMR	`float`	0.005	Base exchange maintenance threshold.
`liquidation_fee_rate`	f_{liq}	`float`	0.0015	Liquidation clearance fee and insurance penalty.
`slippage_rate`	f_{slip}	`float`	0.001	Adverse order book execution slippage buffer.
`margin_mode`	—	`str`	`"linear"`	Margin settlement: `"linear"` or `"inverse"`.
`B`	B	`float`	—	Analytical absorbing liquidation barrier price.
`sigma`	σ	`float`	—	Annualized price volatility of the asset.
`T_days`	T_{days}	`float`	—	Trade holding horizon in days.
`mu`	μ	`float`	0.0	Expected annualized drift rate.
`steps_per_day`	K_{day}	`int`	24	Daily time-step discretization density.
`enable_stochastic_vol`	—	`bool`	`True`	Toggle for Heston stochastic volatility process.
`heston_kappa`	κ_v	`float`	3.0	Mean-reversion speed of the variance process.
`heston_vol_of_vol`	σ_v	`float`	0.35	Volatility of the variance process.
`heston_price_vol_corr`	ρ_{Sv}	`float`	-0.60	Correlation between price shocks and volatility.
`annual_funding_rate`	$\bar{r}_{\text{bull}}$	`float`	0.0	Baseline bull-regime mean funding rate.
`funding_bear_rate`	$\bar{r}_{\text{bear}}$	`float`	-0.05	Negative mean funding rate during bear panics.
`funding_ou_theta`	θ	`float`	5.0	Mean-reversion speed of funding rates.
`funding_ou_sigma`	σ_r	`float`	0.04	Annualized volatility of funding rates.
`funding_price_corr`	ρ_{Sf}	`float`	0.65	Correlation between price shocks and funding.
`num_paths`	M	`int`	5000	Total number of Monte Carlo simulated paths.
`enable_jumps`	1_{jump}	`bool`	`True`	Toggle for Merton Poisson jump component.
`enable_brownian_bridge`	—	`bool`	`True`	Toggle for intra-step continuous barrier detection.
`max_ruin_prob`	α_{max}	`float`	0.01	Maximum allowable probability of ruin (1%).
`use_half_kelly`	c	`bool`	`True`	Fractional Kelly multiplier ($c = 0.5$).
`apply_shrinkage`	—	`bool`	`False`	Toggle to guard against duplicate Bayesian shrinkage.

11 Summary of Practical Risk Guidelines

1. **The Double Penalty of Coin-Margined Trades:** In inverse contracts, collateral loses purchasing power as price declines. As Equations (37) and (40) demonstrate, coin-margined positions experience significantly higher liquidation probability than linear contracts at the same leverage.
2. **Volatility Is Not Constant:** Applying Heston's model proves that market crashes trigger immediate volatility spikes. Risk models that assume fixed volatility underestimate liquidation risk during drawdowns.
3. **Use Half-Kelly Constrained by Ruin Probability:** Theoretical unconstrained Kelly sizing is too aggressive for leveraged derivatives. Always cap leverage using the numerical bisection ruin limit ($\alpha_{\text{max}} \leq 1\%$) to protect capital against total ruin.